

$$\begin{aligned}
\mathbb{E}_{\mathcal{I}, \mathcal{A}} [\|\nabla_x f_{y-c}^{\mathcal{I}}(x)\|_2 \cdot \cos(\alpha_c^{\mathcal{I}, \mathcal{A}})] &= \mathbb{E}_{\mathcal{I}, \mathcal{A}} \left[\|\nabla_x f_{y-c}^{\mathcal{I}}(x)\|_2 \cdot \frac{\langle -\nabla_x f_{y-c}^{\mathcal{I}}(x), \delta^{\mathcal{A}} \rangle}{\|\nabla_x f_{y-c}^{\mathcal{I}}(x)\|_2 \cdot \|\delta^{\mathcal{A}}\|_2} \right] \\
&= \mathbb{E}_{\mathcal{I}, \mathcal{A}} \left[\langle -\nabla_x f_{y-c}^{\mathcal{I}}(x), \frac{\delta^{\mathcal{A}}}{\|\delta^{\mathcal{A}}\|_2} \rangle \right] = \sum_{i=1}^p \mathbb{E}_{\mathcal{I}} [-\nabla_{x_i} f_{y-c}^{\mathcal{I}}(x)] \cdot \mathbb{E}_{\mathcal{A}} \left[\frac{\delta_i^{\mathcal{A}}}{\|\delta^{\mathcal{A}}\|_2} \right],
\end{aligned}$$

where the last equation holds due to the independence of $\delta^{\mathcal{A}}$ and $-\nabla_x f_{y-c}^{\mathcal{I}}(x)$. The expectation of $\frac{\delta^{\mathcal{A}}}{\|\delta^{\mathcal{A}}\|_2}$ does not depend on the samples taken during inference and the expectation of the estimate of the derivative w.r.t. the i -th input $\mathbb{E}_{\mathcal{I}} [-\nabla_{x_i} f_{y-c}^{\mathcal{I}}(x)]$ is the same for different amounts of samples. Therefore, the expectation of the denominator does not change when changing the sample size during inference. This observation explains why the robustness of a stochastic classifier does not depend on the amount of samples taken during inference and gives an justification for picking an arbitrary sample size that allows for a good trade-off between efficiency and reduction of the variance of the MC estimate used for prediction.

5 Experimental robustness analysis

In this section we empirically demonstrate that the findings of our theoretical analysis are transferable to SNNs and help to explain the mechanisms leveraging the experimentally observed robustness of previously proposed SNNs.

Experimental setup Our experiments are conducted on two different image datasets: FashionMNIST [Xiao et al., 2017] and CIFAR10 [Krizhevsky et al., a].⁶ For experiments on FashionMNIST we used feedforward neural networks (FNN) with two stochastic hidden layers, each with 128 neurons. We trained the FNN as a Variational Matrix Gaussian (VMG, the BNN proposed by Louizos and Welling [2016]) via variational inference or as an infinite mixture (IM) with the maximum likelihood objective proposed by Däubener and Fischer [2020], with matrix variate normal distribution placed over the weights. We also trained FNNs of the same architecture, where we added Gaussian noise with $\sigma^2 = \{0.05, 0.1\}$ to the input, by minimizing the cross-entropy. We refer to these models as *stochastic input networks* (SINs) 0.05 and SIN 0.1, respectively. Note, that these networks correspond to the basic networks proposed for randomized smoothing by Lécuyer et al. [2019]. For experiments on CIFAR10 we trained two wide residual networks (ResNet) with MC dropout layers [Gal and Ghahramani, 2016] applied after the convolution blocks and dropout probabilities $p = 0.3$ and $p = 0.6$. If not specified otherwise we used 100 samples of $p(\Theta)$ for inference on all datasets and calculated adversarial attacks with the fast gradient (sign) method (FGM) [Goodfellow et al., 2015], the cross-entropy loss for IMs, BNNs, and ResNets, the margin loss $\mathcal{L}_{\text{margin}}$ specified in section 3 for SINs, and the ℓ_2 -norm constraint based on the CleverHans repository [Papernot et al., 2018]. All experiments were run on a single NVIDIA GeForce RTX 2080 Ti. We refer the reader to supplement B for more details on the datasets, models, and the training procedure.

Accuracy of robustness conditions We first investigated the practical transferability of the derived theorems. For enforcing the smoothness condition used in theorem 4.2, we built on the result from Yang et al. [2022], who showed that the L -smoothness parameter of a classifier $g : x \rightarrow \mathbb{E}_{\epsilon}[f(x + \epsilon)]$ smoothed with random noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$, is bounded by $L \leq 2/\sigma^2$. We therefore applied randomized smoothing during training of the models. For the BNN, on which

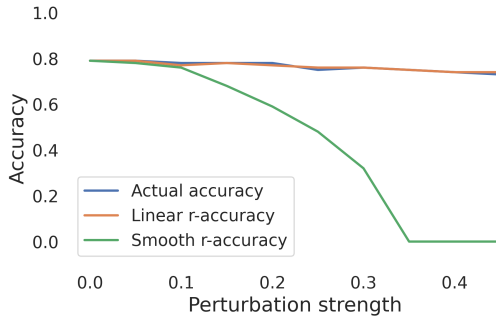


Figure 3: Adversarial accuracy of the smoothed BNN model for attacks based on 10 samples vs percentage of images with $\min_c r_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ (smooth) and $\min_c \tilde{r}_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ (linear) for the first 100 images from the FashionMNIST dataset.

⁶Additional experiments for CIFAR100 are presented in the supplement C.5.

we focus in this section due to space restrictions (results for the other networks look qualitatively similar and can be found in supplement C), we replaced each image in the batch by two noisy copies with Gaussian noise $\epsilon \sim \mathcal{N}(0, 0.1)$ which ensures $L \leq 20$. During prediction we estimated the expectation under Gaussian noise with 10 samples and also used 10 samples for calculating the FGM attack. We estimated the percentage of resulting attacks for which $\min_c r_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ and $\min_c \tilde{r}_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ and compared this to the adversarial accuracy (i.e. the percentage of perturbed samples classified correctly) in figure 3. The percentage of samples fulfilling the condition $\min_c r_c^{\mathcal{I}} > \|\delta^{\mathcal{A}}\|_2$ approaches zero with growing perturbation strength, indicating that the lower bound provided by $r_c^{\mathcal{I}}$ is rather loose due to the high (upper bound of the) L-smoothness constant. On the other hand the percentage of samples for which (3) is fulfilled is a good approximation of the real accuracy for small attack length, which suggests that the discriminant functions are approximately linear in a small neighborhood of the input.

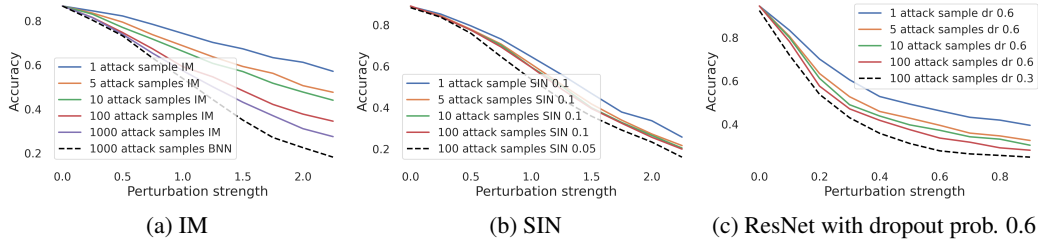


Figure 4: Accuracy under attack for different perturbation strength and amount of samples used for calculating the attack. The dashed line shows the adversarial accuracy for the models with the same architecture but less prediction variance.

Stronger attacks by increased sample size In this section we investigate the effect of varying the amount of samples used for calculating the attack, e.g. for $S^{\mathcal{A}} \in \{1, 5, 10, 100, (1000)\}$. Figure 4 shows the resulting adversarial robustness of the IM, SIN 0.1, and the ResNet with dropout probability 0.6. The accuracy under attack decreases for all models with increasing amount of samples used for calculating the attack as conjectured. We found this to hold also for stronger attacks, i.e. attacks based on logits for IM and BNN (where we observe highly confident softmax predictions), attacks with L_∞ constraint, or *projected gradient descent* (PGD) [Madry et al., 2018] attacks with 100 iterations as shown in supplement C. Note, that iterative attacks already increase the sample size due to their iterative nature, and therefore only few samples per iteration may sufficiently increase attack strength.

Simultaneously to the accuracy we estimated the corresponding values of $\cos(\alpha_j^{\mathcal{I}, \mathcal{A}})$, for $j = \arg \min_c \tilde{r}_c^{\mathcal{I}}$, for the first 1,000 test images and depicted them in figure 5. It can be seen that the cosine values are increasing (which corresponds to decreasing angles) with growing sample size and that the larger the observed values, the lower the accuracy under attack as shown in figure 4. This observation is in accordance to our theoretical analysis which predicts that an increased amount of samples leads to higher cosine values and in turn to less adversarial robustness. Note however, that even when taking many samples $\cos(\alpha_j^{\mathcal{I}, \mathcal{A}}) < 1$, which underlines the fact that the optimal attack direction can not be recovered due to the still existing stochasticity in the inference procedure. Further results on the angle under different amounts of samples during attack can be found in supplement C.2.

Prediction variance as robustness indicator In this section we compare the properties of SNNs that have the same network architecture and a similar training procedure but different prediction variances. That is, we compare the BNN against the IM, SIN 0.05 against SIN 0.1, and the two ResNets with different dropout probabilities against each other. First, we estimated the standard deviation of the prediction and the average standard deviation of the gradient entries of the models by calculating the average of the empirical estimates over the first 1,000 examples from the respective test sets (see table 1). As expected, the IM, SIN 0.1, and ResNet with dropout probability 0.6 have a larger standard deviation of the prediction and hence prediction variance than their respective counterpart. The higher standard deviation of the prediction translates to an increased standard deviation of the gradient. This explains why we observe smaller cosine values for these models compared to their counterparts (see right most boxplots in figure 5 a) and c)) which in turn translate

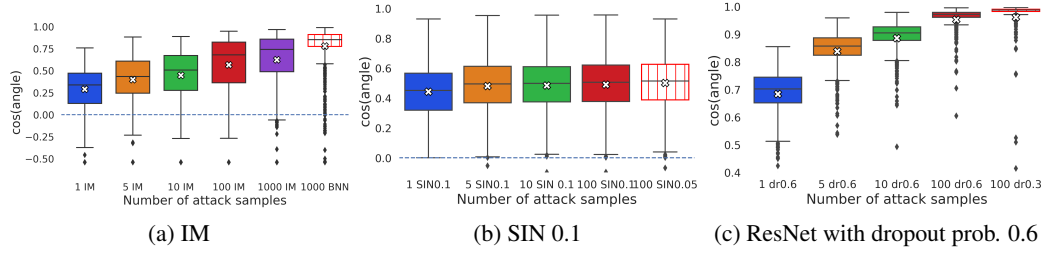


Figure 5: Cosine values for adversarial examples with perturbation strength 1.5 for a) and b) and 0.3 for c) and different amounts of samples. White crosses indicate mean values.

Table 1: Empirical standard deviation of the prediction (for the correct class) based on 1,000 single predictions (each with 1 inference sample) and average length and variance of the corresponding gradient. We report averages over the first 1,000 images from the respective test set.

	FASHIONMNIST				CIFAR10	
	BNN	IM	SIN 0.05	SIN 0.1	DR 0.3	DR 0.6
AVG. STD OF PREDICTIONS	0.0186	0.0473	85.4657	186.5599	0.0146	0.0196
AVG. GRADIENT LENGTH	0.5218	0.5044	49.0120	48.8312	0.0713	0.0599
AVG. STD OF GRADIENT	0.0316	0.0959	1.5741	1.8323	0.0000	0.0000

to an increased robustness (as indicated by the dotted accuracy curves in figure 4). We also found that the length of the mean of the gradient that we estimated based on 1,000 samples is smaller for the models with higher variance.⁷ This might be another reason for the observed smaller angles as discussed in the previous section.

Robustness in dependence of the amount of samples used during inference In practice the amount of samples $S^{\mathcal{I}}$ drawn during inference is fixed to an arbitrary number. In this section we investigate the impact of varying $S^{\mathcal{I}} \in \{1, 5, 10, 100\}$ which are values frequently used. First, we observed that only few samples are necessary to get reliable predictions on clean data for all models as shown by the test accuracies in table 2. Second, we found that increasing the amount of samples did not affect the adversarial accuracy. This can be explained by the observation that the decrease of $\alpha_c^{\mathcal{I}, \mathcal{A}}$ caused by increasing the sample set is counterbalanced by an simultaneous decrease of the average norm of the gradient estimate, as can be seen by inspecting the results in figure 6. That is, the product $\|\nabla_x f_{y-c}^{\mathcal{I}}\|_2 \cdot \cos(\alpha_c^{\mathcal{I}, \mathcal{A}})$ stays approximately the same regardless of the inference sample size as predicted by our theoretical analysis. Interestingly, the effect of increasing the number of samples during inference has almost no effect on the prediction margin (more results are shown in supplement C.4).

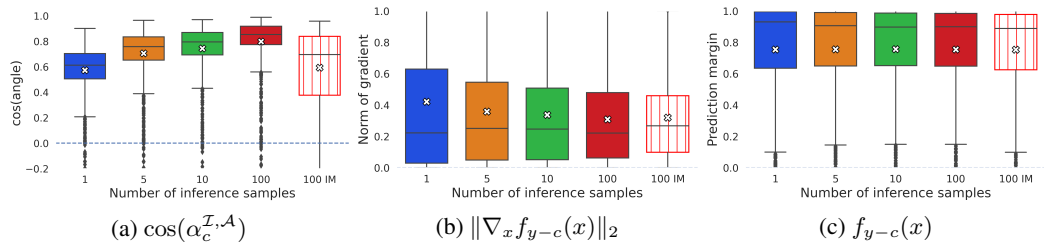


Figure 6: Values of single factors of $\tilde{r}_c^{\mathcal{I}}$ in dependence of varying number of samples used during inference for the BNN trained on FashionMNIST.

⁷For SInS, it is known that a higher variance used during training relates to a lower Lipschitz-constant [e.g. Yang et al., 2022, Salman et al., 2019] which leads to a stronger smoothing effect.